

Maurice Murphy

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SUMMARY

AWS-certified Senior Software Engineer with 6+ years of technical experience building scalable web applications, cloud infrastructure, and data-driven solutions at NBCUniversal and leading tech companies. Specialized in React/TypeScript frontend development, AWS cloud architecture, and business intelligence analytics. Proven expertise in microservices, CI/CD automation, performance optimization, and transforming complex data into actionable insights. Strong technical leader who mentors teams, establishes best practices, and delivers measurable improvements in application performance and business outcomes.

TECHNICAL SKILLS

Frontend Development

Languages & Frameworks: JavaScript (ES6+), TypeScript, React, Redux/Redux Toolkit, React Hooks, Context API

UI/UX Tools: Figma, Balsamiq, Adobe Photoshop, Illustrator

Testing: Jest, React Testing Library, Selenium

Build Tools: Webpack, Vite, npm/yarn

Backend Development

Languages & Frameworks: Node.js (Express), Golang (Gin), Python (Flask, Pandas, NumPy)

API Development: RESTful APIs, Microservices Architecture, JWT Authentication

API Documentation: Swagger/OpenAPI

Cloud & DevOps

AWS Services: EC2, Lambda, API Gateway, S3, CloudFront, RDS, Route 53, DynamoDB, Cognito

Infrastructure as Code: Terraform

CI/CD: GitHub Actions, Automated Testing Pipelines

Containerization: Docker

Version Control: Git, GitHub

Database Management

SQL: MySQL, AWS RDS, PostgreSQL

NoSQL: MongoDB, AWS DynamoDB, Mongoose ODM

Optimization: Query optimization, indexing strategies, schema design

Data Analytics & Visualization

Analysis Tools: Python (Pandas, NumPy, Matplotlib, Seaborn)
Business Intelligence: Tableau, Power BI
Data Processing: ETL pipelines, data cleaning, statistical analysis
Machine Learning: scikit-learn, TensorFlow (AI/ML postgraduate studies)

PROFESSIONAL EXPERIENCE

NBCUniversal
Senior Software Engineer

London, UK | June 2022 – Present

- ▶ **Performance Optimization:** Reduced application load times by 40% through implementing code-splitting, lazy loading, and React performance optimization techniques across high-traffic streaming platform
- ▶ **Frontend Architecture:** Led React/TypeScript development for video streaming features, implementing advanced hooks, Context API, and Redux state management patterns serving millions of users
- ▶ **Cloud Infrastructure & DevOps:** Architected and managed AWS infrastructure using Terraform (Lambda, API Gateway, RDS, CloudFront, S3), reducing deployment time by 50% through automated CI/CD pipelines with GitHub Actions
- ▶ **Microservices Development:** Built scalable backend services in Golang (Gin framework), designing RESTful APIs and event-driven architectures that improved system reliability and response times
- ▶ **Testing & Quality Assurance:** Established comprehensive testing framework using Jest and React Testing Library, achieving 85% code coverage and reducing production bugs by 60%
- ▶ **UI/UX Excellence:** Collaborated with UX designers to transform Figma prototypes into pixel-perfect, accessible (WCAG 2.1 AA compliant) interfaces with seamless cross-device experiences
- ▶ **Technical Leadership:** Mentored 3 junior developers, conducted code reviews, and established TypeScript coding standards and best practices that improved team velocity by 25%
- ▶ **AWS Integration:** Implemented secure authentication flows using AWS Cognito and optimized data storage solutions with DynamoDB and RDS for scalable backend operations

Fifty One Degrees

London, UK | November 2021 – May 2022

Data Analyst

- ▶ **Business Intelligence:** Designed and implemented interactive dashboards using Tableau and Power BI, providing real-time insights to C-suite executives that informed strategic decisions and improved operational efficiency by 30%
- ▶ **Data Analysis & Reporting:** Conducted exploratory data analysis using Python (Pandas, NumPy, Matplotlib) to identify trends, patterns, and anomalies across customer behavior, product performance, and revenue streams
- ▶ **SQL Optimization:** Wrote complex SQL queries and optimized database performance for PostgreSQL and MySQL, reducing report generation time by 50%
- ▶ **Data Visualization:** Created compelling data visualizations and executive reports that translated complex datasets into actionable business insights for non-technical stakeholders
- ▶ **KPI Development:** Established key performance indicators and automated reporting processes that tracked business metrics, enabling data-driven decision-making across product and marketing teams
- ▶ **Cross-Functional Collaboration:** Partnered with product, engineering, and marketing teams to define data requirements, validate hypotheses, and deliver analytics solutions that drove product improvements

Freelance Software Engineer

London, UK | November 2020 – October 2021

Independent Contractor

- ▶ **Full-Stack Delivery:** Built and deployed 8+ production web applications using React, Node.js/Express, and MongoDB for clients in e-commerce, healthcare, and fintech sectors
- ▶ **Client Solutions:** Translated business requirements into technical specifications, delivering projects on-time and within budget using Agile methodologies and weekly sprint cycles
- ▶ **API Development:** Designed and implemented RESTful APIs with JWT authentication, rate limiting, and comprehensive Swagger documentation, handling 10K+ daily requests
- ▶ **Database Architecture:** Engineered MongoDB database schemas with optimized indexing strategies, improving query performance by 70%
- ▶ **Quality Assurance:** Implemented unit and integration testing with Jest and Selenium, maintaining 80%+ code coverage across all projects
- ▶ **Modern Stack Expertise:** Leveraged React Hooks, Redux Toolkit, Express middleware, and Mongoose ODM to build maintainable, scalable applications

- **Python Automation:** Engineered Python-based automation tools that streamlined building plan generation, reducing manual drafting time by 60% and eliminating human error in calculations
- **Data Analytics:** Developed data processing pipelines using Python (Pandas, NumPy) for project cost estimation and sustainable design analysis across £100M+ projects
- **Energy Simulation Tools:** Created automated building energy simulation systems that accelerated eco-friendly architectural decision-making by 40%
- **Workflow Optimization:** Integrated Python automation into Revit BIM workflows using PyRevit, improving design accuracy and reducing project delivery timelines by 3 weeks
- **Technical Mentorship:** Trained 5+ junior technologists in Python automation and scripting best practices, establishing reusable code libraries

EDUCATION

National College of Ireland

Postgraduate Diploma in Computer Science – Artificial Intelligence/Machine Learning (Part-time)

Dublin, Ireland | January 2021 – January 2022

Code Institute

Diploma in Full Stack Software Engineering (Part-time)

Dublin, Ireland | September 2020 – August 2021

University of Westminster

BSc (Honours) in Architectural Technology (2:1)

London, UK | September 2013 – July 2016

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- AWS Certified Developer – Associate
- Google Cybersecurity Certificate

KEY ACHIEVEMENTS

- ▶ **Performance Impact:** Reduced application load times by 40% at NBCUniversal through React optimization, serving millions of streaming platform users
- ▶ **Cloud Architecture:** Designed and deployed AWS infrastructure with Terraform, cutting deployment cycles by 50% and improving system reliability
- ▶ **Quality Engineering:** Established testing frameworks achieving 85% code coverage, reducing production bugs by 60%
- ▶ **Career Transition:** Successfully pivoted from architectural technology to software engineering, leveraging Python automation expertise
- ▶ **Independent Delivery:** Built and launched 8+ production applications as freelancer, maintaining 100% client satisfaction
- ▶ **Technical Leadership:** Mentored junior developers and established coding standards that improved team velocity by 25%
- ▶ **AWS Certified:** Earned AWS Cloud Practitioner and Developer Associate certifications, demonstrating cloud architecture expertise